

NANCY-ESSEY AP, France

WMO: 071800

Lat: 48.6878N

Lon: 6.2214E

Elev: 751

StdP: 14.30

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	21.6	11.7	10.3	19.9	15.9	12.7	23.5	24.0	46.2	21.5	45.9	4.7	40	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	89.1	68.2	85.0	67.1	81.6	66.0	70.2	84.2	68.7	81.5	67.2	78.7	7.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	98.2	74.7	64.4	93.2	73.1	63.0	88.5	71.9	34.5	84.1	33.3	81.6	32.1	78.9	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.9	16.5	14.4	DB	11.5	94.5	5.4	3.9	7.6	97.3	4.5	99.6	1.4	101.8	-2.5	104.7
(5)				WB	10.8	73.4	5.3	1.6	7.0	74.5	3.9	75.4	0.9	76.3	-2.9	77.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	36.4	38.3	44.0	49.9	57.3	63.8	67.6	66.7	59.4	52.3	43.5	38.0
	DBStd	12.84	8.11	7.46	6.60	6.30	6.11	6.25	5.72	5.80	5.44	6.61	6.93	7.78
	HDD50	1689	425	329	204	79	9	0	0	0	1	54	213	375
	HDD65	5232	887	747	652	453	250	97	37	48	183	396	646	836
	CDD50	2239	2	2	17	76	235	415	546	518	283	124	17	4
	CDD65	307	0	0	0	0	10	62	118	101	15	1	0	0
	CDH74	3369	0	0	0	18	164	700	1274	1028	177	8	0	0
	CDH80	1111	0	0	0	0	30	215	459	375	32	0	0	0
Wind	WSAvg	6.4	7.1	7.4	7.2	6.6	6.3	6.2	6.0	5.4	5.7	5.8	6.2	7.2
Precipitation	PrecAvg	30.2	2.4	2.3	2.3	2.0	2.7	2.7	2.5	2.6	2.6	2.5	2.8	2.9
	PrecMax	38.7	6.1	6.3	6.0	5.4	7.8	7.0	6.8	5.5	5.3	7.4	7.3	7.9
	PrecMin	18.5	0.4	0.3	0.2	0.4	0.4	0.3	0.3	0.2	0.2	0.1	0.6	0.2
	PrecStd	5.2	1.3	1.5	1.3	1.3	1.2	1.5	1.3	1.4	1.4	1.6	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	59.3	68.4	76.4	84.0	91.8	94.8	96.0	84.1	75.0	62.9	56.8
		MCWB	51.7	49.2	54.0	60.0	66.8	69.1	69.8	69.1	65.7	62.2	56.2	51.6
	2%	DB	52.8	55.0	63.6	72.2	78.3	86.0	88.9	88.4	79.2	69.9	58.8	53.9
		MCWB	49.3	48.0	52.0	57.5	63.2	67.8	68.3	68.3	64.5	60.9	53.9	50.3
	5%	DB	50.2	52.0	59.2	67.4	74.5	81.6	85.2	83.7	74.7	66.0	56.1	51.6
		MCWB	47.1	47.0	50.2	55.3	61.3	66.6	67.1	66.6	62.9	59.2	52.4	48.7
	10%	DB	47.6	49.2	55.2	63.3	70.6	77.7	81.4	79.7	70.7	62.4	53.5	49.0
		MCWB	44.8	45.3	48.8	53.2	59.3	64.6	65.7	65.5	60.9	57.4	50.6	46.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	50.9	55.8	62.0	67.9	71.8	72.4	71.9	68.3	64.2	57.6	53.0
		MCDB	54.9	56.2	64.5	74.2	80.4	85.6	87.6	86.6	79.0	71.6	61.1	55.3
	2%	WB	49.9	49.2	53.3	58.5	65.2	69.4	70.1	69.9	65.9	61.9	54.7	50.9
		MCDB	52.4	53.6	60.3	69.6	75.8	81.8	84.7	84.2	76.4	67.9	57.9	53.6
	5%	WB	47.2	47.4	51.4	56.0	62.5	67.5	68.4	68.3	64.1	59.8	52.8	48.8
		MCDB	49.8	51.3	57.7	65.5	71.9	79.0	81.6	80.5	72.7	64.8	55.8	51.4
	10%	WB	45.0	45.5	49.4	53.9	60.4	65.8	66.8	66.6	62.2	57.9	50.6	46.7
		MCDB	47.4	48.9	54.9	61.8	68.6	76.3	78.6	77.0	69.0	62.2	53.2	48.9
Mean Daily Temperature Range	5% DB	MDBR	9.6	12.5	16.8	19.6	19.5	20.5	20.2	20.1	18.8	15.7	10.6	9.0
		MCDBR	11.5	17.9	25.6	28.8	27.1	27.9	28.2	28.7	26.3	21.1	14.2	11.2
		MCWBR	9.1	12.2	15.5	16.0	13.7	12.5	10.7	11.6	13.2	12.8	10.1	8.7
	5% WB	MCDBR	10.8	15.1	20.9	25.5	24.4	25.5	25.1	25.4	22.7	18.6	13.1	10.7
		MCWBR	9.0	11.0	13.4	14.7	13.0	12.3	10.2	11.3	12.1	11.8	9.9	8.6
Clear-Sky Solar Irradiance	taub	0.311	0.335	0.362	0.392	0.402	0.407	0.402	0.399	0.373	0.358	0.328	0.308	
	taud	2.437	2.384	2.321	2.255	2.265	2.284	2.290	2.325	2.372	2.405	2.434	2.421	
	Ebn at Noon	239	258	269	271	272	271	271	266	262	246	229	223	
	Edh at Noon	21	27	34	40	41	41	40	37	32	26	21	18	
All-Sky Solar Radiation	RadAvg	285	524	911	1364	1613	1835	1722	1489	1139	674	330	229	
	RadStd	44	91	136	181	180	159	168	155	115	53	33	40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page